

**In one line: The device is the attacker's:** everything in the .ipa is readable and every client check is hookable — so secrets and real decisions live on the **server**, and the client only *proves* things (TLS, App Attest). Privacy is the mirror image: you get **one** system prompt per permission, and must **declare** what you touch (purpose strings, privacy manifest, nutrition label).

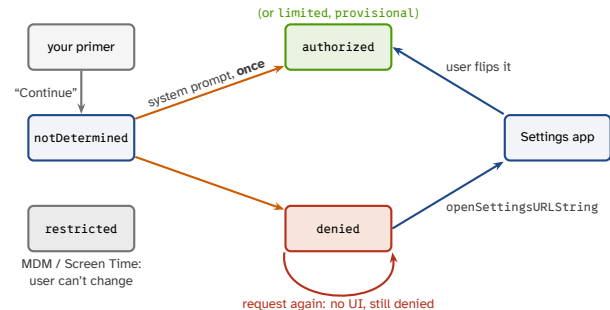
## Hardening

- **ATS:** URLSession requires HTTPS, TLS 1.2+, forward secrecy, SHA-256 certs. Exceptions in Info.plist  
NSAppTransportSecurity: per-domain  
NSExceptionDomains (prefer), NSAllowsLocalNetworking;  
blanket NSAllowsArbitraryLoads needs an App Review justification.
- **Pinning:** ATS trusts every CA on the device (user/MDM CAS too); pin the **SPKI hash** + a **backup pin** (NSPinnedDomains or the trust challenge) — see certificate-pinning-2026.
- **No secrets in the binary:** strings, class-dump, Hopper read plists and literals; an XOR'd key is plaintext in memory at use (Frida). Client keys: low-privilege, scoped, rotatable; real secrets stay behind your backend.
- **Jailbreak detection** (Cydia paths, fork(), injected dylibs) runs on the attacker's device ⇒ hookable. A risk *signal*, never a boundary. Same for obfuscation and PT\_DENY\_ATTACH: they buy time, not safety.
- **App Attest** (DCAppAttestService): SE key → attestKey (Apple-signed: genuine app, real device) → generateAssertion per sensitive request; the **server** verifies.
- **DeviceCheck** (DCDevice): 2 bits per device per developer, survive reinstall — “free trial used?”, not integrity.
- **Storage recap:** secrets → Keychain; files → Data Protection; never UserDefaults; pasteboard .localOnly + .expirationDate; blur the app-switcher snapshot.
- **Logging:** Logger/os\_log redact **dynamic strings** as <private> by default (scalars are public) (*unverified*); opt in per value with privacy: .public. Never print() a token.

## Privacy

- **Purpose string** per protected API (NSCameraUsageDescription, NSLocationWhenInUseUsageDescription, ...): missing ⇒ **crash** on first access. Specific and honest.
- Read status first (authorizationStatus); **ask in context**, after your own primer, since the system prompt is one-shot.
- **ATT:** ATTrackingManager.requestTrackingAuthorization (+ NSUserTrackingUsageDescription, app active). Not authorized ⇒ IDFA is all zeros. “Tracking” = linking your data with *other companies* for ads/brokers. Fingerprinting is banned either way.
- **Privacy manifest** PrivacyInfo.xcprivacy (app and each SDK): NSPrivacyTracking, NSPrivacyTrackingDomains (blocked without ATT consent), NSPrivacyCollectedDataTypes, NSPrivacyAccessedAPITypes + reason codes.  
**Required-reason APIs:** UserDefaults (CA92.1 (*unverified*)), file timestamps, boot time, disk space, active keyboards. Listed SDKs must ship a manifest + signature.
- **Nutrition label** (App Store Connect): data collected, linked to identity, used to track — *including your SDKs*.

## One prompt, then Settings



## Example — log safely, branch on status

```

let log = Logger(subsystem: "com.x.app", category: "auth")
log.info("login \(email)") // shows <private>
log.info("screen \(name, privacy: .public)") // explicit opt-in
log.info("user \(uid, privacy: .private(mask: .hash))")

switch AVCaptureDevice.authorizationStatus(for: .video) {
case .notDetermined: // primer first,
    showPrimer { requestCameraAccess() } // then the ONE prompt
case .denied:        openSettings() // can't re-ask
case .restricted:    explainUnavailable() // user can't change
case .authorized:    startCamera()
@unknown default: break
}
  
```

## Traps

- ! “Re-ask after Don’t Allow” — impossible; the request returns denied silently. Explain + deep link to Settings.
- ! Camera permission ≠ ATT: using data in-app is not “tracking”.
- ! Jailbreak check or evaluatePolicy as the gate — both are Booleans on a hostile device.
- ! “Our API key is obfuscated” — still public; scope and rotate it.
- ! NSAllowsArbitraryLoads to fix one HTTP host — use a per-domain exception.
- ! UserDefaults needs a required-reason entry — even in a plain app.

## Remember

**“Readable, hookable, declared.”** Everything shipped is readable; every check is hookable; every access is declared. Trust → server + attestation. Permissions: **primer** → **one prompt** → **Settings**.